THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 10: **Tissues in Action** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Grasses grow again after being cut. Reason (R): Intercalary meristem is present at the base of their leaves. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A ring of bark is removed from a branch. After some days, the part above the ring swells due to: 1

(a) accumulation of water **(b)** minerals
(c) accumulation of food **(d)** gas

3. The tissue that produces the enzymes and hormones in glands is: 1

(a) squamous epithelium **(b)** glandular epithelium
(c) muscle **(d)** bone

4. The reflex action of removing the hand from a hot object involves: 1

(a) blood only
(b) sensory neuron, relay neuron and motor neuron
(c) muscle only
(d) bone only

5. A plant tissue is stained with safranin and some cells turn red. They contain: 1

(a) lignin **(b)** cellulose only **(c)** starch **(d)** protein

6. Assertion (A): The trunk of a tree becomes thicker every year. Reason (R): Cambium divides and adds secondary xylem and phloem. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. After removing a ring of bark, the leaves above remain fresh for some time because: 1

(a) cork is intact **(b)** xylem is intact
(c) cambium is intact **(d)** phloem is intact

8. A smooth muscle cell has: 1

(a) branching **(b)** striations
(c) a single nucleus and spindle shape **(d)** many nuclei

9. The impulse in a neuron travels from: 1

(a) axon to dendrite **(b)** cell body to dendrite
(c) axon to cell body **(d)** dendrite to axon

10. Which of these stains is used for a section of a stem to show the xylem? 1

(a) iodine **(b)** methylene blue **(c)** eosin **(d)** safranin

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain why wood is made mainly of xylem and why it is strong. 2

12. State two differences between the movement of materials in xylem and phloem. 2

13. What is the effect on a plant if all meristems are destroyed? 2

14. How would you show that xylem conducts water in a plant? State the observation. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A potato disc and a carrot disc are placed in iodine solution. Predict the observation and identify the tissue involved. 3

16. State the reasons for the huge numbers of mitochondria in the cardiac muscle cells. 3

17. State the role of the vascular cambium and the cork cambium in a woody stem with their products. 3

OR

A student observed a section showing many xylem vessels in a ring and phloem outside it with a cambium layer between. Identify the plant organ and describe the arrangement. 3

18. Explain how a tree grows in height and thickness over many years, mentioning each meristem. 3

OR

Explain how the shape of the stomata changes with the water status of guard cells in the day and the night. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Design an experiment to show that water moves upward through the xylem and food moves downward through the phloem in a plant (using red ink and girdling) and describe the observations and conclusions. 5

OR

Explain the importance of structural adaptations of tissues for survival of plants in different habitats, with examples from xerophytes, hydrophytes and mesophytes. 5

20. Describe the structure of the dicot stem and the monocot stem, and the differences in the arrangement of the vascular bundles and the cambium. 5

OR

Compare tissue regeneration in plants and animals, giving examples of tissues with high, low and no regeneration, and discuss applications in medicine. 5

END OF QUESTION PAPER — Total: 40 Marks